

Day 32 – AI Screening API & End-to-End System Integration

1. Objective

Day 32 focuses on validating the existing Zecpath AI System through its REST API and end-to-end ATS demonstration.

The objective is to verify that the existing API layer can expose ATS functionality including resume processing, candidate scoring, candidate shortlisting, health monitoring, and asynchronous job handling.

No new architecture was introduced during this evaluation. The existing project structure and components were used.

2. Existing API Implementation

The API implementation is located at:

ats_engine/api.py

The system uses FastAPI to expose the ATS functionality through REST endpoints.

The API application is configured as:

- Application: Zecpath ATS API
- Version: 1.0.0
- Base path: /api/v1

3. API Endpoints

The existing API provides the following endpoints:

Method	Endpoint	Purpose
GET	/api/v1/health	Check API availability
POST	/api/v1/resumes/upload	Upload a candidate resume
POST	/api/v1/resumes/parse	Process resume text
POST	/api/v1/scoring	Generate ATS candidate score
POST	/api/v1/shortlisting	Rank and shortlist candidates
GET	/api/v1/jobs/{job_id}	Retrieve asynchronous job status

4. Health Check

The health endpoint verifies that the ATS API is operational.

Endpoint:

GET /api/v1/health

The endpoint returns a standardized successful response containing:

- Service name
- Service status

The existing API uses the status:

healthy

5. Resume Upload

The resume upload endpoint accepts candidate resume files.

Supported file types:

- PDF
- DOCX

The endpoint validates:

- File extension
- Empty file uploads

A successful upload creates:

- Resume ID
- Asynchronous job ID
- QUEUED processing status

The existing API uses FastAPI BackgroundTasks for the demonstration asynchronous processing workflow.

6. Resume Parsing

The resume parsing endpoint accepts resume text through a validated request schema.

Endpoint:

POST /api/v1/resumes/parse

The existing implementation performs basic text normalization and returns parsing information.

The API also provides an integration point where the existing resume parser pipeline can be connected.

7. ATS Candidate Scoring

The scoring endpoint uses the existing ATSScoringEngine.

Endpoint:

POST /api/v1/scoring

The scoring request contains:

- Job role
- Skill match
- Experience relevance
- Education alignment
- Semantic similarity

All score values are constrained between 0 and 100.

The existing ATS scoring engine calculates the final candidate score using these evaluation components.

8. Candidate Shortlisting

The shortlisting endpoint uses the existing CandidateShortlistingEngine.

Endpoint:

POST /api/v1/shortlisting

Candidate information includes:

- Candidate ID
- Candidate name
- Candidate score

The existing engine ranks candidates and determines their shortlisting decisions.

9. Asynchronous Job Handling

The API contains an in-memory job store for the current implementation.

Supported job states are:

- QUEUED
- PROCESSING
- COMPLETED
- FAILED

The job status endpoint allows the client to retrieve the current processing state.

Endpoint:

GET /api/v1/jobs/{job_id}

The current implementation is intended for API design and testing. Persistent storage can be integrated in a production deployment.

10. Standard API Responses

The API uses standardized response structures.

Successful responses contain:

- success

- message
- data

Error responses contain:

- success
- error
 - code
 - message

This provides a consistent response contract across API operations.

11. API Contract Configuration

The existing API contract is stored in:

data/api_contracts.json

The configuration defines:

- API version
- Base path
- Available endpoints
- Job statuses
- Error codes

The configured API version is:

v1

The base path is:

/api/v1

12. End-to-End ATS Demonstration

The existing ATS demonstration was executed using:

```
python -m tests.demo_ats_system
```

The demonstration evaluated three candidates for the target role:

Data Scientist

Required skills included:

- Python
- SQL
- Machine Learning
- Pandas
- Scikit-learn

13. Candidate Evaluation Results

Candidate A

- Candidate ID: DEMO001
- Profile Type: Strong Match
- Skill Match: 100.0%
- Experience Relevance: 90.0%
- Education Alignment: 90.0%
- Semantic Similarity: 92.0%
- Overall ATS Score: 94.6
- Recommendation: Excellent Candidate
- Decision: Shortlisted

Candidate B

- Candidate ID: DEMO002
- Profile Type: Moderate Match
- Skill Match: 60.0%
- Experience Relevance: 72.0%
- Education Alignment: 75.0%
- Semantic Similarity: 76.0%
- Overall ATS Score: 68.7
- Recommendation: Average Candidate
- Decision: Review

Candidate C

- Candidate ID: DEMO003
- Profile Type: Low Match
- Skill Match: 0.0%
- Experience Relevance: 40.0%
- Education Alignment: 65.0%
- Semantic Similarity: 35.0%
- Overall ATS Score: 25.0
- Recommendation: Needs Improvement
- Decision: Rejected

14. Ranking and Shortlisting Summary

Rank	Candidate	ATS Score	Decision
1	Candidate A	94.6	Shortlisted
2	Candidate B	68.7	Review
3	Candidate C	25.0	Rejected

Overall demonstration result:

- Total candidates: 3
- Shortlisted: 1

- Review: 1
- Rejected: 1

The end-to-end ATS demonstration completed successfully.

15. API Test Validation

The existing API test suite was executed using:

```
python -m pytest tests/test_ats_api.py -q
```

Result:

6 passed

One existing warning was reported:

StarletteDeprecationWarning

The warning concerns the use of httpx with the Starlette test client and did not cause any test failure.

16. System Integration Validation

The Day 32 evaluation confirms that the existing components can operate together through the ATS workflow.

The demonstrated flow includes:

Resume / Candidate Data ↓ ATS Evaluation ↓ Skill Match ↓ Experience Relevance ↓ Education Alignment ↓
Semantic Similarity ↓ Overall ATS Score ↓ Candidate Ranking ↓ Shortlisting Decision

The API layer provides an interface for exposing these capabilities to backend or client applications.

17. Main System Entry Point

The project also contains:

main.py

The current main entry point initializes the project environment and application logging.

It displays:

Welcome to Zecpath AI System

The main application also records startup and environment initialization events through the existing logger.

18. Existing Architecture

Day 32 continues to use the existing modular architecture.

Relevant components include:

- ats_engine
- parsers

- scoring
- screening_ai
- utils
- tests
- data
- docs

The API layer integrates with existing scoring and shortlisting components instead of replacing them.

19. Test Results

The Day 32 validation produced the following results:

Evaluation	Result
End-to-end ATS demo	Passed
API test suite	6 passed
Candidate ranking	Passed
Candidate shortlisting	Passed
ATS scoring	Passed
API response handling	Passed
Asynchronous job design	Validated

20. Warning Observed

One existing warning was reported during API testing.

Starlette/httpx Deprecation Warning

The current environment reports a deprecation warning related to the use of httpx with the Starlette test client.

This warning does not affect the successful API test result.

No warning-related code change was required for the Day 32 validation.

21. Day 32 Deliverable

Day 32 establishes an end-to-end validation of the existing Zecpath ATS API and screening workflow.

The evaluation confirms that:

- The FastAPI layer is operational.
- Existing ATS scoring functionality is accessible through the API.
- Candidate shortlisting is accessible through the API.
- API request and response contracts are defined.
- Asynchronous job handling is implemented.

- The end-to-end ATS demonstration completes successfully.
- Candidate ranking and shortlisting produce expected demonstration results.
- The API test suite passes successfully.

No architectural replacement was performed during Day 32.

22. Final Status

Day 32 API and end-to-end integration validation completed successfully.

API tests:

6 passed

End-to-end ATS demonstration:

Completed successfully

The existing system is ready to proceed to the next development and evaluation stage.